Well, you see, gradient descent (or a variation on the theme) is the optimization technique that is used behind the scenes by many machine learning algorithms like the one mentioned above. But did you know that, long before being used as the back-bone of supervised learning, this technique was designed to solve optimization problems, like some of the ones we have seen on graphs?

If you didn't, or if you'd like to delve into this topic and learn more about how gradient descent works, this should be the perfect chapter for you.

Remember that in chapter 15 we have discovered that while for planar graphs there are efficient algorithms that find a planar embedding in linear time, not all graphs are planar, and it's not always possible to draw a graph's edges in the plane without them intersecting?

What's worse, each non-planar graph has a minimum number of intersections when drawn in the plane, but there is not (and there cannot be[1]) any efficient algorithm that can find this number (or an embedding with as few intersections as possible) for any generic graph.

The minimum edge crossing problem is, in fact, NP-hard, and as of today, verifying crossing numbers is unfeasible for many categories of non-planar graphs with more than ~20 vertices.